SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x20=40)

Q.25 Design a socket and spigot. (CO4,CO6)

Q.26 a) A 70mm diameter solid shaft is welded to a flat by a fillet weld around the circumference of the shaft. Determine the size of the weld if the torque on the shaft in 4kNm. The maximum shear stress in the weld is 80MPa. (CO2, CO4, CO6)

b) Design a riveted lap joint for the 6mm thick plate.

Q.27 Design and draw a cast iron coupling for a mild steel shaft transmitting 100kW at 300rpm. The allowable shear stress in the shaft is 60 Mpa and the angle of twist is not to exceed 1^{o} in a length of 20 diameters. The allowable shear stress in the coupling bolt is 30 Mpa.

Roll No. ....

5th Sem / Mech
Subject:- Machine Design

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Endurance limit of the material can improve by (CO1)

a) Coating b) Shot peening
c) Heat treatment d) Polishing

Q.2 The shear stress on the principle plane is (CO1)

a) Zero b) Minimum
c) Maximum d) None of the above

Q.3 Rankine's theory of failure is applicable in case of (CO4)

a) Ductile materials b) Brittle materials s
c) Elastic materials d) Plastic materials

Q.4 Which of the following key transmits power through frictional resistance only? (CO6)

a) Saddle key b) Feather key
c) Woodruff key d) Sunk key

Q.5 What is the material of eye end and fork end of the knuckle joint? (CO3)

a) Cast iron b) Copper

c) Aluminium d) Mild steel

Q.6 Transverse fillet welded joints are designed for (CO6)

a) Tensile strength

b) Compressive strength

c) Bending strength

d) Torsion strength

Q.7 The holes in the plates for riveting is usually made by (CO6)

a) Drilling b) Punching

c) Arc cutting d) Gas cutting

Q.8 In a flange coupling, the weakest element should be (CO6)

a) Flange b) Key

c) The bushes d) Shaft

Q.9 Slope of the thread is (CO1)

a) Half of the pitch b) Double of the pitch

c) Thrice of the pitch d) One fourth of the pitch

Q.10 Rivets are made of (CO3)

a) Brittle material b) Ductile material

c) Soft material d) Hard material

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x2=20)

Q.11 Define volumetric strain. (CO1)

Q.12 Define working stress. (CO1)

Q.13 Name the various types of shafts. (CO5)

Q.14 Define principal planes. (CO1)

Q.15 Define equivalent twisting moment. (CO1)

Q.16 What is function of a key? (CO6)

Q.17 Write the use of knuckle joint. (CO6)

Q.18 What is the function of knuckle joint? (CO6)

Q.19 Define butt weld joint. (CO6)

Q.20 Write the function of power screw. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any three questions out of four questions. (10x3=30)

Q.21 Explain the characteristics of a good designer. (CO1)

Q.22 Explain the maximum shear stress and maximum principle strain theories. (CO4)

Q.23 Define a shaft when subjected to combined twisting and bending moment. (CO5)

Q.24 A rectangular sunk key 18mm wide, 12mm thick and 90mm long is required to transmit a torque 26KN-m from a 100mm diameter shaft. Calculate the induced shear and crushing stresses in the key. (CO6)